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PALEOMAGNETIC AND RADIOMETRIC AGE DETERMINATION OF (PERMIAN) PEGMATITES IN THE MIDDLETOWN DISTRICT (CONNECTICUT)

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Presently available paleomagnetic poles for the Permian of North America are derived mainly from sediments. The occurrence of radiometrically dated pegmatites in Connecticut with sufficient magnetic mineral contents provided a possibility of obtaining paleomagnetic poles for a more narrow time interval in the Permian. The new poles fall essentially in the same cluster as those obtained from redbeds in West Virginia and eastern Canada.

Emplacement of the pegmatites (258 ± 1 my) occurred in a period of tectogenesis and thermal activity that may have marked the initial phase of the Permo-Triassic eastdrift of North America.

1. Introduction

Numerous pegmatites are exposed in the crystalline highlands near Middletown, Connecticut. Their petrology and geologic setting have been described in great detail by Cameron et al. [1] and Stugard [2]. The pegmatites are believed to have been emplaced in the Late Permian by magmatic injection [3]. Two types of Late Permian pegmatites are noted: (a) conformably intercalated sill-like bodies which are relatively flat, lens-shaped and greatly elongated in the direction of strike; and (b) less common, large, elliptical bodies. The longest dimensions of both types generally parallel the N–S trend of the regional tectonics. In addition, they are commonly zoned and usually possess a complex mineralogy and chemistry [1, 2].

Small, stringer-type, folded, quartzo-feldspathic pegmatites are also common but are apparently unrelated to the larger types in that they are significantly older (350 my) [4] and possess a simple chemistry and mineralogy. These will not be discussed further here.

The pegmatites studied are steeply W-dipping masses most often emplaced in the Brimfield schist, close to the Brimfield schist–Glastonbury gneiss contact. The contact zone evidently offered less resistance to injection than the relatively dense Glastonbury gneiss. Forceful injection of the pegmatites has been argued [1, 5] due to minor fold development in their host rocks as well as local, cross-cutting pegmatite-host rock contacts. The crushing, distortion and crumpling of the host rocks adjacent to the pegmatites can be attributed to pre-emplacement faulting or post-emplacement folding. In the latter case the pegmatites acted as rigid masses around which the schists were draped and folded. Some of the isolated lenses of the Strickland quarry pegmatites can be explained by boudinage of the brittle masses during post-magmatic folding. In most cases the pegmatite-host rock contacts are extremely sharp.

2. Radiometric studies

Numerous radiometric dates have been determined by various methods, most of which are summarized in Brookins [6]. He reports the following results: (a) U,

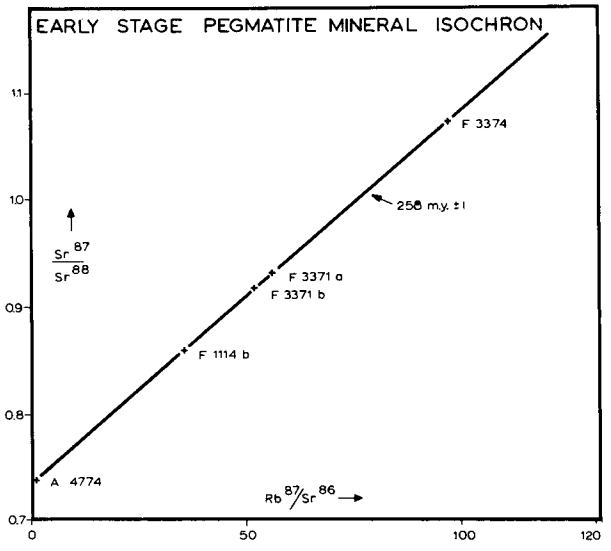


Fig. 1. Early stage mineral isochron for the pegmatites of the Middletown district, modified from Brookins et al. [3]. F = K-feldspar, A = Plagioclase, $\lambda Rb87 = 1{,}39 \times 10^{-11}$/yr. Initial $^{87}Sr/^{86}Sr = 0.734 \pm 0.002$.

Th-Pb and Pb-Pb ages on uraninites and samarskites: 261 ± 3 my ($\bar{\sigma}$); (b) K-Ar ages for muscovites and lepidolites: 250 ± 2 my ($\bar{\sigma}$); (c) Rb-Sr mineral isochron (fig. 1): 258 ± 1 my ($\bar{\sigma}$). These data are remarkably concordant and argue for crystallization at about 255 my ago. Some K-Ar and Rb-Sr dates from host rock micas, however, are as young as 220 to 230 my [7]. The young ages are thought to be due to diffusion of radiogenic Ar and Sr, due in turn to relatively small grain size of the host rock micas. Methot [8], however, has found that whole rock samples from at least two pegmatites (Hale quarry, White Rocks quarry) yield pre-250 my dates for the fine grained aplite, but dates as young as 225 my for coarse grained pegmatites. The difference may reflect the cooling time of the pegmatites or the inability to adequately obtain a whole rock sample from the coarse grained material. The young dates suggest that thermal events continued into the Triassic. Late stage cleavelandites (An 0-3) contain excess radiogenic Sr and have not been used in construction of the isochron shown in fig. 1.

It is noteworthy that the pegmatite mineral isochron (fig. 1) has been constructed from samples from several different pegmatites, which may be an argument for a common parent source (a view also entertained by Lundgren [9] based on entirely different considerations). If various sources for the pegmatitic material were available then one might expect a series of parallel mineral isochrons, the initial ratio of each reflecting its particular source. This is not observed and the one-parent magma hypothesis is adhered to. The host rocks for the pegmatites are not thought to be the source for the pegmatites as their initial ratios and Sr contents relative to those of the pegmatites are so different as to preclude any genetic relationship [3, 7].

3. Paleomagnetic studies

Most Permian poles for North America are derived from measurements on clastic sediments. The age of these sediments is known only approximately and estimates range from Permo-Carboniferous to Permo-Triassic. To estimate the location of the magnetic pole during the lower Permian, the remanent magnetic directions of two of the radiometrically dated pegmatites were measured. The Strickland and Hale pegmatites were chosen because they belong to a relatively small group of pegmatites with measurable remanent magnetization. Ilmenomagnetite occurs in the contact zone of the Strickland pegmatite and hematite is found dispersed throughout the Hale pegmatite.

Fine grained ilmeno-magnetite (Ti/Fe ratio of separates, 0.27) occurs as schlierens parallel to the foliation of the schist in the western contact zone of the Strickland pegmatite and is probably derived from the Brimfield schist by assimilation.

Hematite occurs throughout the Hale pegmatite in three different 'forms'. Most dominant are schlieren of hematite coated feldspars which occur predominantly near gneissic xenoliths. Locally, red bands occur, composed of granular hematite, that follow fine crack systems that may have been pathways for late magmatic solutions. The graphic pegmatite zone in the center of the body contains numerous small disseminated red patches. Several of these patches have a yellow aureole that surrounds small columbite crystals. A late magmatic alternation of this iron-rich niobium oxide could explain the occurrence of the hematite in the central portion of the pegmatite.

Thirty samples were cored from both the Strickland and the Hale quarry. In each case two or more sites were chosen at lateral distances of approximately 50 m.

Because of the high quartz content, considerable difficulty was experienced with drilling the samples in the field; therefore, most cores were drilled in the laboratory from oriented samples. The resulting reorientation might have increased error. Specimens were measured with a spinner magnetometer and were cleaned in A.C. fields (peak value 900 Oe).

The intensity of the NMR ranged from 1.4×10^{-3} to 5.1×10^{-5} for the samples with ilmeno-magnetite and from 3.7×10^{-4} to 4×10^{-6} for the samples containing hematite. Of the 60 samples, however, 14 had to be discarded because of insufficient magnetization in the early stages of demagnetization. Although the cores were sufficiently long, usually only one specimen could be used per core for the same reason.

The samples containing ilmeno-magnetite were generally clean after having been subjected to A.C. fields ranging from 100 to 200 Oe. Samples with hematite required higher fields, ranging from 500 to 700 Oe which is rather surprising because its stability in rocks such as redbeds is usually much greater.

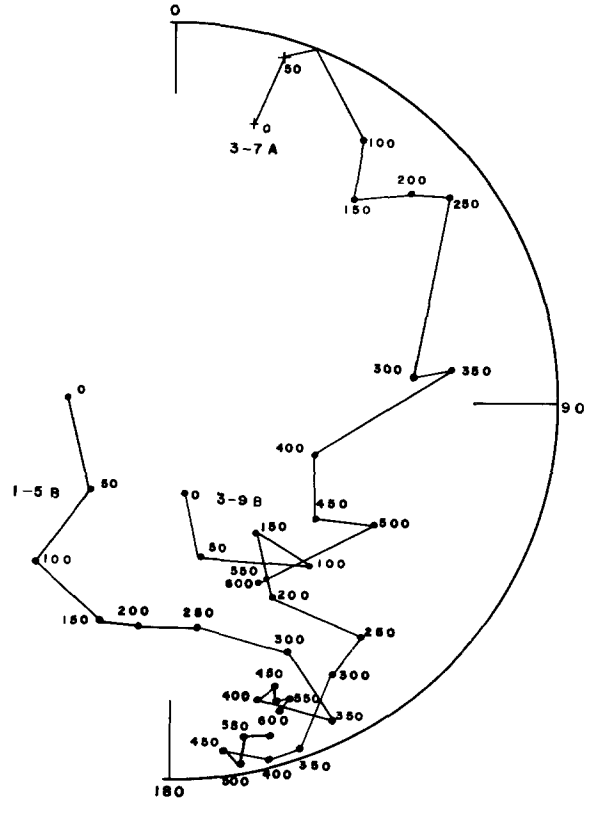


Fig. 2. Directional change of remanent magnetization during A.C. demagnetization for 3 samples from the Hale pegmatites. Numbers in Oe. + inclination negative. • inclination positive.

Fig. 2 shows the change in magnetic direction during demagnetization for samples 1–5b, 3–7a and 3–9b. Rapid inclination changes occurred in the lower A.C. fields; in intermediate fields the declination changes appear more dominant. Of interest is sample 3–7a, the NRM of which exhibited reversed characteristics. Demagnetization caused the remanent magnetic direction to relate over some 140° toward the cluster of directions obtained from the other samples. Further demagnetization rendered the samples too weak to measure.

The declinations for the site averages range from 160 to 174; the inclinations from +11 to +34 (table 1). The field that existed at the time of the magnetization was reversed and present day Connecticut was located south of the magnetic equator. The virtual magnetic poles for the pegmatites (fig. 3) have a latitudinal variation from 30° to 41° N and a longitudinal variation from 124° to 144° E. The latter variation is rather large and is caused mainly by the deviating remanent direction of site 5. It is possible that this direction was somewhat modified by post-emplacement microfolding of the contact zone. Although this folding is not visible in the sampling area, some crumpling exists some 10 m westward. Because the fold axis of these folds is more or less coaxial with the mean declination of the other sites, the folding cannot have influenced the remanent direction significantly. The other four pegmatite poles fall within a cluster of poles obtained from the redbeds of the Dunkard formation (West Virginia) [10], the Hurley Creek formation (New Brunswick) [11] and the Prince Edward Island (P.E.I.), Pictou (Nova Scotia) and Bonaventure formations (Quebec) [12]. The congruence of these poles, derived from igneous rocks and sediments, reaffirms the Permian age of the pegmatites and suggests that the inclination error in the redbeds is insignificant. Whether or not deposition of the redbeds was contemporaneous with emplacement of the pegmatites cannot be established so far because there may have been a period of relative polar stability in the Permian.

The Paleozoic magnetic poles for eastern North America occur primarily in a relatively narrow latitudinal band (30° to 40°N) that extends from about

Table 1
Paleomagnetic results for two pegmatites from the Middletown district.

	Site	c/s	Before demag.		After demag.		α_{95}	Pole			
			D	*I*	*D*	*I*		Lat N	Long E	$d\varphi$	$d\chi$
Hale Quarry											
41°37'45" N	1	7/9	21	+64	167	+16	4	39	124	3	5
72°35'45" W	2	12/14	336	+39	165	+11	7	41	127	5	8
	3	6/9	56	+72	160	+23	6	33	131	6	8
Strickland Quarry											
41°35'30" N	4	6/10	78	+57	159	+21	6	33	132	3	6
72°35'30" W	5	11/11	110	+43	174	+34	9	30	144	5	9

C/s = core specimen; *D*, *I* = declination, inclination; α_{95} = semi-angle of cone of confidence at 95% level.

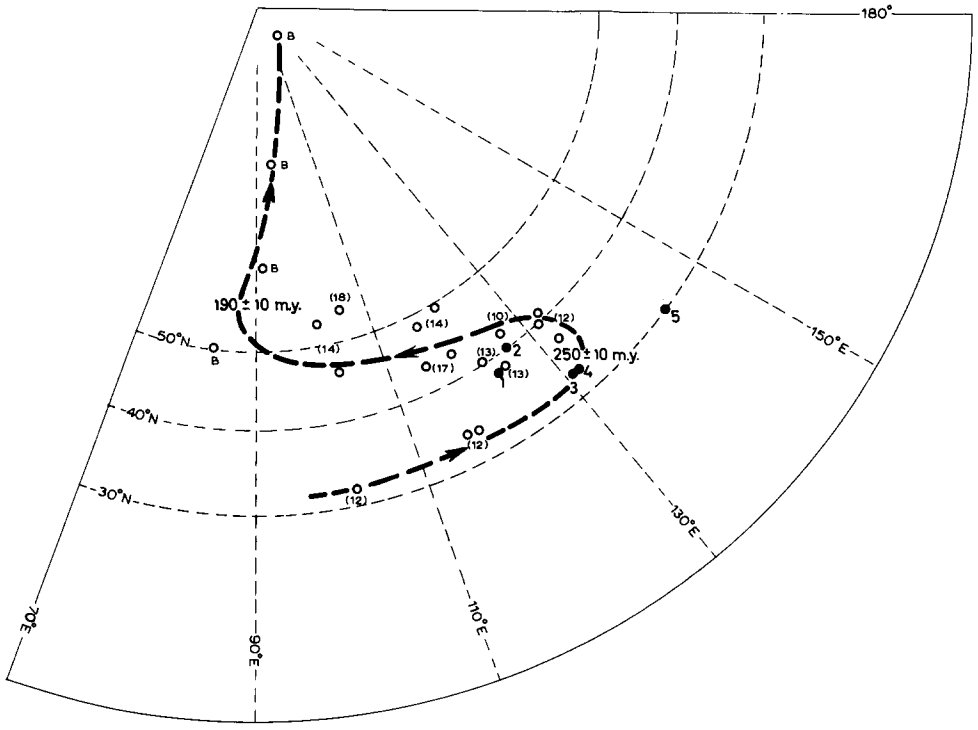


Fig. 3. Plot of virtual magnetic poles from the pegmatites of the Middletown district [1–5]. Added, a series of mean poles, provided by refs. [10–14, 17, 18] and De Boer (B). The dashed line represents a hypothetical polar path for the late Paleozoic and early Mesozoic.

100° E to 135° E [13]. Presently the most western of these poles belong to the late Silurian Bloomsburg formation; the most eastern belong to the early Permian, Prince Edward Island redbeds. If we assume that these sediments obtained their chemical remanent magnetization penecontemporaneously with deposition, then the polar shift can be estimated to have been ± 3 cm/yr. This relatively low rate suggests that the shift may have resulted primarily from continental drift and that actual polar motion was insignificant. The eastward motion of the poles was reversed in the Permian. Recent information on redbeds from the west [14] and the position of the late Triassic poles suggest that during the Permian and early Triassic the magnetic pole moved westward from approximately 135° to 90°E in a latitudinal band between 40° and 50°N. If we consider our information on the age and pole positions of the Permian pegmatites (this paper) and the Triassic lava-flows [15], then it appears that this motion may have occurred at a minimal rate of 7 cm/yr. In late Triassic time this pattern was changed again and polar motion became primarily longitudinal with the rate of shift increasing to 20 cm/yr or more [15].

Although insufficient information is available at this time, it appears that the following phenomena combined to provide the late Paleozoic-early Mesozoic polar path:

(a) Northward longitudinal motion of the magnetic pole that originated in Permian time and accelerated to a maximum in the late Triassic, before diminishing in the early Jurassic;

(b) Pre-Permian latitudinal motion of the magnetic pole in an eastward direction, caused by west drift of Laurasia and subsequent Permo-Triassic latitudinal motion of the pole in westward direction, caused by east drift of North America. The reversal in the drift pattern probably initiated the thermal and dynamic events of New England in the period 230 to 250 my [16], during which the pegmatites were emplaced.

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